UCRL

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$$\max_{\pi_0,\dots,\pi_{H-1}} \sum_{t=0}^{H-1} E\left(r(s_t, a_t)\right)$$

- H: horizon
- The optimal policy may no longer be stationary, it may be time dependent, hence $\pi_t(a_t|s_t)$
- for simplicity assume $r(s,a)$ is deterministic
- Recall how to solve such finite-horizon MDPs using the dynamic programming backward recursion:

$$V_H^*(s) = 0 \quad \forall s$$

$$V_t^*(s) = \max_a \; r(s,a) + E\left(V_{t+1}(s_{t+1}) \,\middle|\, \substack{s_t=s \\ a_t=a}\right)$$

- optimal value function is $V_0^*(s_0)$. And the optimal policy is deterministic.
- In RL, the goal is to solve the MDP without knowing $P(\cdot|\cdot,\cdot)$
- Each episode consists of running the MDP from 0 to H-1. We repeat the episodes each time with a new policy.
- Let t denote the tth episode started from some initial condition s_0^t.
- The goal is to achieve small regret:

$$\sum_{t=1}^{T} \left(V_0^*(s_0^t) - V_0^{\pi^t}(s_0^t)\right),$$

where π^t is the policy used in the tth episode: $\pi^t = (\pi_0^t, \dots, \pi_{H-1}^t)$

UCRL algorithm:

$n_{s,a}(t) = \max(\#$ times (s,a) was visited in the first t episodes, $1)$

$n_{s,a,s'}(t) =$ # times the MDP transitioned from s to s' when action a was taken in state s

$\Rightarrow$

$$\hat{P}_t(s'|s,a) = \frac{n_{s,a,s'}(t)}{1 \vee n_{s,a}(t)} \qquad (a \vee b \text{ means } \max(a,b))$$

In episode $t+1$, do the following:

- Compute a confidence set $C_{t,\delta}$ around $\hat{P}_t$
- Use the policy $\pi^{t+1} = (\pi_0^{t+1}, \dots, \pi_{H-1}^{t+1})$ s.t.

$$\pi^{t+1} = \arg\max_{\pi} \max_{P \in C_{t,\delta}} V_0(s_0^{t+1})$$

Note: Similarity to MAB. Optimistically choose the best policy within the confidence bound. "Optimism in the face of uncertainty".

- Later, we will specify how $C_{t,\delta}$ is to be chosen.
- Let $\tilde{P}^t$ be the maximizing prob. transition matrix in the algorithm

$$V_0^*(s_0^{t+1}) - V_0^{\pi_{t+1}}(s_0^{t+1})$$

$$= V_0^*(s_0^{t+1}) - \tilde{V}_0(s_0^{t+1}) + \tilde{V}_0^{\pi_{t+1}}(s_0^{t+1})$$

$$= V_0^{\tilde{}}\left(s_0^{t+1}\right) - \tilde{V}_0\left(s_0^{t+1}\right) + \tilde{V}_0^{\tilde{\pi}_{t+1}}\left(s_0^{t+1}\right) - V_0^{\pi_{t+1}}\left(s_0^{t+1}\right)$$

$\tilde{V}_0$ is the value function if the prob. trans. matrix was $\tilde{P}$ while the $V_0^{\tilde{}}$ and $V_0^{\pi_{t+1}}$ are under the true P^*.

$$\leq \underbrace{\tilde{V}_0^{\tilde{\pi}_{t+1}}\left(s_0^{t+1}\right) - V_0^{\pi_{t+1}}\left(s_0^{t+1}\right)}_{\triangleq A_0^{t+1}}$$

this would be true if $C_{t,s}$ includes the true P^* since $\tilde{P}$ is chosen to the best prob. trans. matrix in $C_{t,s}$ that maximizes the reward

$$= \cancel{r\left(s_0^{t+1}, a_0^{t+1}\right)} + \sum_{s'} \tilde{P}\left(s' \mid s_0^{t+1}, a_0^{t+1}\right) \tilde{V}_1^{\tilde{\pi}_{t+1}}(s')$$

$$- \cancel{r\left(s_0^{t+1}, a_0^{t+1}\right)} - \sum_{s'} P^*\left(s' \mid s_0^{t+1}, a_0^{t+1}\right) V_1^{\pi_{t+1}}(s')$$

$$= \sum_{s'} \tilde{P}\left(s' \mid s_0^{t+1}, a_0^{t+1}\right) \tilde{V}_1^{\tilde{\pi}_{t+1}}(s')$$

$$- \sum_{s'} P^*\left(s' \mid s_0^{t+1}, a_0^{t+1}\right) \tilde{V}_1^{\tilde{\pi}_{t+1}}(s')$$

$$+ \sum_{s'} P^*\left(s' \mid s_0^{t+1}, a_0^{t+1}\right) \tilde{V}_1^{\tilde{\pi}_{t+1}}(s')$$

$$- \sum_{s'} P^*\left(s' \mid s_0^{t+1}, a_0^{t+1}\right) V_1^{\pi_{t+1}}(s')$$

$$= \sum_{s'} \left(\tilde{P}\left(s' \mid s_0^{t+1}, a_0^{t+1}\right) - P^*\left(s' \mid s_0^{t+1}, a_0^{t+1}\right)\right) \tilde{V}_1^{\tilde{\pi}_{t+1}}(s')$$

$$+ \sum P^*\left(s' \mid s_0^{t+1}, a_0^{t+1}\right) \left(\tilde{V}_1^{\tilde{\pi}_{t+1}}(s') - V_1^{\pi_{t+1}}(s')\right)$$

$$+ \sum_{s'} P^*(s' \mid s_0^{t+1}, a_0^{t+1}) \left(\tilde{V}_1^{\pi_{t+1}}(s') - V_1^{\pi_{t+1}}(s') \right)$$

$$\leq H \left\| \tilde{P}(\cdot \mid s_0^{t+1}, a_0^{t+1}) - P^*(\cdot \mid s_0^{t+1}, a_0^{t+1}) \right\|_1 \qquad \left(\text{Assume } r \in [0,1] \right)$$

$$+ \sum_{s'} P^*(s' \mid s_0^{t+1}, a_0^{t+1}) \, \Delta_1^{t+1}(s')$$

$\Rightarrow$

$$\Delta_0^{t+1}(s_0^{t+1}) \leq 2H f_\delta\left(n_{s_0^{t+1}, a_0^{t+1}(t)}\right) + \sum_{s'} P(s' \mid s_0^{t+1}, a_0^{t+1}) \, \Delta_1^{t+1}(s')$$

because $\| P' - \hat{P}_t \|_1 \leq \| \hat{P}_t - \tilde{P} \|_1$

w.p. $\geq 1-\delta$

If $C_{t,\delta}$ is defined as follows:

$$C_{t,\delta} = \left\{ P : \left\| P(\cdot \mid s,a) - \hat{P}_t(\cdot \mid s,a) \right\|_1 \leq f_\delta\left(n_{s,a}^{(t)}\right) \right\} \quad \forall s,a$$

and $f_\delta\left(n_{s,a}^{(t)}\right)$ is chosen s.t.

$$P^* \in C_{t,\delta} \quad \text{w.p.} \geq 1-\delta \quad \forall t$$

Let $F_{t,h}$ be the history containing

$$\left\{ (s_0', a_0'), \ldots, (s_{H-1}', a_{H-1}'), \ldots \right.$$
$$\left. (s_0^t, a_0^t), \ldots (s_h^t, a_h^t) \right\}$$

and Δ_0^{t+1} and Δ_1^{t+1} be random variables taking values determined by s_0^{t+1}, s'.

Then,

$$\Delta_0^{t+1} \leq 2H f_\delta\left(n_{s_0^{t+1}, a_0^{t+1}}^{(t)}\right) + E\left(\Delta_1^{t+1} \mid F_{t+1,0}\right)$$

In general,

$$\Delta_h^{t+1} \le 2H\, f_\delta\left(n_{s_h^{t+1}, a_h^{t+1}}(t)\right) + E\left(\Delta_{h+1}^{t+1} \mid F_{t+1,h}\right)$$

with $\Delta_H^{t+1} = 0$

To facilitate telescoping, define

$$\eta_h^t = \Delta_h^t - E\left(\Delta_h^t \mid F_{t,h-1}\right)$$

$$\Rightarrow \Delta_h^{t+1} \le 2H\, f_\delta\left(n_{s_h^{t+1}, a_h^{t+1}}(t)\right) + \Delta_{h+1}^{t+1} - \eta_{h+1}^{t+1}$$

with $\Delta_A^{t+1} = 0$

$$\Rightarrow \Delta_h^{t+1} - \Delta_{h+1}^{t+1} \le 2H\, f_\delta\left(n_{s_h^{t+1}, a_h^{t+1}}(t)\right) - \eta_{h+1}^{t+1}$$

$$\Rightarrow \boxed{\Delta_0^{t+1} \le 2H \sum_{h=0}^{H-1} f_\delta\left(n_{s_h^{t+1}, a_h^{t+1}}(t)\right) - \sum_{h=0}^{H-1} \eta_{h+1}^{t+1}}$$

We will choose

$$f_\delta(n) = \sqrt{\frac{2\left(S\log 2 + \log\frac{n(n+1)SA}{\delta}\right)}{n}}$$

where S is the # states, A is the # actions in the MDP. The extra SA multiplying $n(n+1)$ in later two boxes where we will

in the MDP. The extra SA multiplying $\frac{n(n+1)}{\delta}$ is for later purposes where we will use a union bound over all (s,a) pairs.

Recall: $P(P^* \in C_{t,\delta}) \geq 1-\delta$ with this choice of f.

Thus, we have proved that w.p. $\geq 1-\delta$,

$$\Delta_0^{t+1} \leq 2H \sum_{h=0}^{H-1} f_\delta\left(n_{s_h^{t+1}, a_h^{t+1}}(t)\right) - \sum_{h=0}^{H-1} \eta_{h+1}^{t+1}$$

where

$$\eta_{h+1}^{t+1} = \Delta_{h+1}^{t+1} - E\left(\Delta_{h+1}^{t+1} \mid F_{t+1,h}\right)$$

and

$$\Delta_h^{t+1} = V_h^{\tilde{\pi}_{t+1}}(s_h) - V^{\pi_{t+1}}(s_h)$$

$\Rightarrow \eta_h^{t+1}$ is a martingale difference sequence, i.e., $E\left(\eta_h^{t+1} \mid F_{t+1,h}\right) = 0$

and $\left|\eta_h^{t+1}\right| \leq 2H$ assuming $r(s,a) \in [0,1]$

$$\text{Regret}_T = \sum_{t=1}^{T} \Delta_0^t$$

$$\leq 2H \sum_{t=1}^{T} \sum_{h=0}^{H-1} f_\delta\left(n_{s_h^t, a_h^t}(t-1)\right)$$

$$- \sum_{t=1}^{T} \sum_{h=0}^{H-1} \eta_0^t .$$

$$-\sum_{t=1}^{T}\sum_{h=0}^{H-1}\eta^{t}_{h+1}$$

By the Azuma-Hoeffding inequality (which states that the Hoeffding bound also applies to martingale difference sequences and not just independ r.v.'s)

$$P\left(\frac{1}{HT}\sum_{t=1}^{T}\sum_{h=0}^{H-1}-\eta^{t}_{h+1} > \varepsilon\right) \leq \exp\left(\frac{-2HT\varepsilon^2}{2H^2}\right)$$

If we want this to be $\leq \delta$,

$$\frac{T\varepsilon^2}{2H} \geq \log\frac{1}{\delta}$$

or $$\varepsilon^2 \geq \frac{2H}{T}\log\frac{1}{\delta}$$

or $$\varepsilon \geq \sqrt{\frac{2H}{T}\log\frac{1}{\delta}}$$

or $$HT\varepsilon \geq HT\sqrt{\frac{2H}{T}\log\left(\frac{1}{\delta}\right)}$$

$$\Rightarrow -\sum_{t=1}^{T}\sum_{h=0}^{H-1}\eta^{t}_{h+1} \leq H\sqrt{2HT\log\left(\frac{1}{\delta}\right)}$$

w.p. $\geq 1-\delta$

Next, we will bound

$$\sum_{t=1}^{T}\sum_{h=0}^{H-1} f_\delta\left(n_{s_h^t, a_h^t}^{(t-1)}\right)$$

$$= \sum_{t=1}^{T}\sum_{h=0}^{H-1} \sqrt{\frac{2\left(S\log 2 + \log\frac{SA\, n_{s_h^t, a_h^t}^{(t-1)}\left(n_{s_h^t, a_h^t}^{(t-1)}+1\right)}{\delta}\right)}{1 \wedge n_{s_h^t, a_h^t}^{(t-1)}}}$$

$$\le \sum_{t=1}^{T}\sum_{h=0}^{H-1}\sum_{s,a} \frac{\mathbb{I}_{s_h^t = s,\, a_h^t = a}}{\sqrt{1 \wedge n_{s,a}^{(t-1)}}} \times \underbrace{\sqrt{2\left(S\log 2 + \log\frac{SA\, HT(HT+1)}{\delta}\right)}}_{\triangleq\; C_{S,A,H,T,\delta}}$$

$$= C_{S,A,T,H,\delta} \sum_{t=1}^{T}\sum_{s,a} \frac{m_{s,a}(t)}{\sqrt{n_{s,a}^{(t-1)}}}$$

$$\le C \sum_{t=1}^{T}\sum_{s,a} \frac{m_{s,a}(t)}{\sqrt{1 \wedge \left(n_{s,a}(t) - H\right)}}.$$

since

$$n_{s,a}(t) = n_{s,a}(t-1) + m_{s,a}(t)$$

and $m_{s,a}(t) \le H$

$$= c \sum_{t=1}^{T} \sum_{s,a} \frac{m_{s,a}(t)\, \mathbb{I}_{n_{s,a}(t) > H}}{\sqrt{n_{s,a}(t) - H}}$$

$$+ c \underbrace{\sum_{t=1}^{T} \sum_{s,a} m_{s,a}(t)\, \mathbb{I}_{n_{s,a}(t) \le H}}$$

$$\underbrace{\sum_{s,a} \underbrace{\sum_{t=1}^{T} m_{s,a}(t)\, \mathbb{I}_{n_{s,a}(t) \le H}}_{\le H}}_{\le SAH}$$

$$\le c \sum_{t=1}^{T} \sum_{s,a} \frac{m_{s,a}(t)\, \mathbb{I}_{n_{s,a}(t) > H}}{\sqrt{n_{s,a}(t) - H}} + c\,SAH$$

$\Rightarrow$ from the algebraic identity in the previous lecture,

$$\le 2c \sum_{s,a} \sqrt{n_{s,a}(T) - H} + cSAH \le 2c \sqrt{n_{s,a}(T)} + cSAH$$

$$= 2cSA \; \frac{1}{SA} \sum_{s,a} \sqrt{n_{s,a}(T)} + cSAH$$

$$\cdots \frac{}{SA}$$

$$\leq 2CSA \sqrt{\frac{1}{SA} \underbrace{\sum_{s,a} n_{s,a}(T)}_{=TH}} + CSAH$$

↑ Jensen

$$= 2C\sqrt{SAHT} + CSAH.$$

Thus, w.p. $\geq 1-2\delta$,

$$\text{Regret}_T \leq 4HC\sqrt{SAHT} + 2CSAH^2 + H\sqrt{2HT\log\left(\frac{1}{\delta}\right)}$$

where

$$C = \sqrt{2\left(S\log 2 + \log \frac{SAHT(HT+1)}{\delta}\right)}$$

Since the max regret $\leq HT$, choosing $\delta = \frac{1}{\sqrt{T}}$

gives

$$E(\text{Regret}_T) \leq 4HC\sqrt{SAHT} + 2CSAH^2 + H\sqrt{2HT\log T}$$

$$+ 2H\sqrt{T}$$

with

$$C = \sqrt{2\left(S\log 2 + \log SAHT^2(HT+1)\right)}.$$

For large T, ignoring log terms,

$$E(\text{Regret}_T) = \tilde{O}\left(H^{3/2} S \sqrt{AT} \right)$$